TELL-ME: Toward Personalized Explanations of Large Language Models

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Abstract

This paper introduces TELL-ME, a prototype designed to provide personalized explanations for outputs generated by large language models (LLMs). Traditional explanation approaches often fail to address diverse user needs in understanding generative AI (GenAI) outputs. TELL-ME bridges this gap by tailoring explanations to three distinct user groups: beginners, advanced users, and experts. Two literature reviews on explanation and personalization informed the design of TELL-ME. With that, TELL-ME provides feature ablation, counterfactual, and self-explainability explanations on text generation tasks. A preliminary evaluation of TELL-ME with six participants showed that the personalized explanations enhanced users' understanding of model outputs. Nonetheless, areas for improvement were identified, such as integrating state-of-the-art LLMs into the prototype. The full code for TELL-ME is available at https://github.com/jayjkb/TELL-ME.

CCS Concepts

• **Human-centered computing** → **User models**; Usability testing; • **Information systems** → **Decision support systems**; • **Computing methodologies** → *Natural language processing*.

Keywords

Personalization, Customization, Explainable AI, GenXAI, Generative AI, Artifact Development

ACM Reference Format:
Jakub Jeck, Florian Leiser, Anne Hüsges, and Ali Sunyaev. 2025. TELL-ME: Toward Personalized Explanations of Large Language Models. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 18 pages. https://doi.org/10.1145/3706599.3719982

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CHI EA '25, Yokohama, Japan
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ACM ISBN 979-8-4007-1395-8/25/04
https://doi.org/10.1145/3706599.3719982

1 Introduction

Interest in generative artificial intelligence (GenAI) surged with the release of widely accessible applications like ChatGPT [52]. GenAI refers to a subset of artificial intelligence (AI) that uses machine learning (ML) techniques to generate outputs–such as text, images, or code–by learning from existing data [16]. However, GenAI models introduce risks of producing incorrect or misleading outputs. For instance, large language models (LLMs) may generate hallucinations that are nonsensical or undesired [29].

To mitigate users' reliance on undesired GenAI outputs, it is crucial to help them understand how these models generate their outputs [53]. GenAI models are billion-parameter models, designed to infer user requests [52]. However, this size makes these models even more opaque to users than traditional "black-box" AI models. As a result, the growing number of GenAI users often lack insight into the reasoning behind the outputs they receive [55]. Explainable AI (XAI) offers techniques that help users understand how AI models associate inputs with outputs [55]. For instance, feature attribution methods can quantify the relevance of specific input components (e.g., a word or pixel), enabling users to identify which aspects of an input contribute most to a generated output [43]. Combining XAI with GenAI, termed generative explainable AI (GenXAI), can offer human-understandable explanations for GenAI outputs [59]. Through GenXAI, the reasoning processes of GenAI models can be elucidated and could enable users to recognize and address erroneous outputs more effectively [59].

With the vast availability of explanation methods, initial approaches investigated and categorized explanation methods, such as feature attribution, for LLMs [43, 82]. However, appropriate explanations require congruence with the needs of the AI user [41]. More effective explanations can be provided when the explanation's purpose and the user's familiarity with XAI are known [41]. Yet, GenAI explanations have been mostly static and non-adjustable, producing one-size-fits-all explanations. Such explanations might fail to meet user needs, being too technical for novice users or overly simplistic for experts, potentially leading to misunderstandings or misuse of AI-generated outputs [79]. Therefore, these one-size-fits-all explanations limit the users' ability to personalize the explanations based on their background, expertise, or needs [59]. While personalizing explanations has been investigated for non-generative AI systems (e.g., [60]), it remains unclear which personalization approaches are suitable for explanation methods of GenAI outputs.

To address this issue, we account for personalizations of explanations in a prototype called `TELL-ME`: Toward personalized explanations of large language models. TELL-ME allows personalization of explanation methods for three distinct user groups (i.e., beginners, advanced users, and experts). TELL-ME integrates explanation methods with personalization through several mechanisms, including feature ablation combined with feature prioritization and presentation adjustments, counterfactual explanations with complexity reduction, and self-explainability enhanced by user-background incorporation, interactivity, and on-demand explanations. Building on existing research on GenXAI, which primarily adapts XAI methods to LLMs (e.g., [43, 82]), TELL-ME extends current LLM architectures rather than creating a novel one. The prototype focuses on text generation as its primary use case. However, sentiment analysis as a second use case highlights its generalizability in implementing personalized explanations. For implementation, TELL-ME utilizes GPT-2 [25], Phi-2 [26] and a fine-tuned version of BERT [24]. A preliminary evaluation with six participants demonstrated that TELL-ME improved their understanding of GenAI reasoning and enhanced their comprehension of GenAI outputs.

2 Prototype Design

During the development process, we identified commonly used personalization and explanation methods, derived key requirements for the prototype, aligned our selection of methods with these requirements, and designed an intuitive, user-friendly interface. The following sections outline the results, beginning with an overview of the personalization and explanation methods used.

2.1 Personalization Methods

Personalization tailors explanations to user needs by adjusting formats (e.g., using numbers or colors) to match expertise or preferences [60]. To explore existing methods, we conducted a literature review based on [72], focusing on *explanation* personalization, due to varying definitions across contexts [15]. Using the search string and criteria detailed in Appendix A, we conducted forward and backward searches, concluding the process on May 12, 2024. This resulted in the identification of 24 articles on personalization approaches for AI explanations, each supported by examples or case studies. Six key concepts for personalizing explanations were identified through thematic analysis [5]. The concept matrix is presented in Appendix B:

User Background Incorporation. Incorporating user background involves leveraging knowledge about the user's expertise, preferences, and mental models to craft tailored and intuitive explanations. User knowledge is crucial, as generic explanations often fail to resonate with users, potentially leading to meaningless explanations [28]. Frameworks addressing this include post-hoc and "selective explanations," inspired by human selective reasoning processes [28, 37]. Similarly, aligning explanations with user mental models enhances usability and interpretability [20, 49]. Research underscores the importance of incorporating user background for explanation interpretability, as no *one-size-fits-all* explanation exists [33, 58, 66].

Presentation Format. Different users prefer different explanation formats, requiring XAI systems to personalize to textual, visual, auditory, or hybrid explanation formats [21]. Studies showed that lay users benefit from textual explanations, while experts prefer visual ones, and both groups achieve optimal results with hybrid formats [66]. Research also highlights the potential of virtual agents, with human-like interaction further enhancing user experience [75].

Interactivity. Interactivity allows users to personalize explanations in an interactive process. Dialogue systems, such as TalkToModel, were introduced to enable users to understand ML models through an LLM-powered dialogue engine [62]. Similarly, the concept of *explanation exploration* has been mentioned, which involves an initial explanation that is subsequently refined through user queries [67]. Further research supports these approaches, stating that effective explanations depend on understanding specific user questions [40].

Explanation on Demand. Explanation on demand enhances user control by providing information only when needed, improving alignment with individual cognitive needs and preventing information overload [11]. Tools, such as TalkToModel, can be seen as approaches to provide explanations on demand through a dialogue system [62]. Users seem to prefer requesting explanations to control their timing and access to supplementary information [7, 21].

Complexity Reduction. Complexity describes the size or number of elements in an explanation and the number of relationships between the concepts presented in the explanation. Higher complexity increases response times and reduces user satisfaction [32, 48]. Users prefer single-line complex explanations over multi-line ones in textual explanations. Other scholars have recommended personalizing explanations by dividing them into three levels of complexity, and progressively increasing detail with each level [6].

Feature Prioritization. Feature prioritization in explanations aims to highlight the most important aspects of a model's decisions. One approach allows users to personalize explanations by selecting feature subsets, leading to faster and more accurate responses to follow-up questions about the AI prediction [38]. Another approach uses heatmaps to visually shift user focus to key features [9].

Prototype Requirements. TELL-ME should support a wide range of personalized explanations. Since the primary purpose of LLMs is to provide accurate predictions, sufficient prediction capabilities stand before all personalization approaches. Therefore, the system must enable users to input text and receive generative model outputs. Additionally, we identified six key requirements for TELL-ME to allow personalized explanations:

- R1: `User Background Incorporation`: Users should be able to specify their level of expertise, preferred explanation methods, and familiarity with AI to receive explanations tailored to their background. By aligning with users' backgrounds, the explanations can better resonate with their experience and understanding.
- R2: `Presentation Format`: Multiple explanation formats (e.g., visual and textual) should be provided to support the user in comparing different explanation methods and find the method best suited to the user's understanding.

Table 1: Mapping of Requirements to XAI Methods

Personalization Approaches	Feature Ablation	Counterfactuals	Self-Explainability
User Background Incorporation			X
Presentation Format	X		
Interactivity			X
Explanation on Demand			X
Complexity Reduction		X	
Feature Prioritization	X	X	

R3: `Interactivity`: The interface should allow users dialogue-based interaction, enabling follow-up questions and deeper exploration of explanations, leveraging the capabilities of existing LMs.

R4: `Explanation on Demand`: Users should control when and what explanation types are presented.

R5: `Complexity Reduction`: Explanations should be presented in one frame to avoid diversion of the users. Additionally, the complexity of the explanation should be adjusted to the users expertise.

R6: `Feature Prioritization`: Related to reduced complexity, users with low expertise should only be presented with the most important features. As user expertise increases, additional features might be presented.

2.2 Explanation Methods

Based on R2, TELL-ME should allow presenting multiple explanations simultaneously, such as graphs and textual summaries. In alignment with R1, R5 and R6, the visualization and detail of the explanations should be adjusted to the user's knowledge level.

To identify suitable explanation methods, we conducted a literature review on existing explanation approaches based on [72], enhanced by forward and backward searches. The process concluded on May 10, 2024, yielding 32 papers that met the search string and inclusion criteria outlined in Appendix A. The approaches were classified across three dimensions: model specificity [63], explanation timing [63], and explanation focus [82]. Our findings comprised 5 explanation methods containing 15 subcategories whose concept matrices are presented in Appendix C. For a detailed overview of explanations of GenAI outputs, we refer to [59].

Feature Attribution Methods. We identified five subcategories for feature attribution methods that quantify the relevance of input elements, like words [43, 59]. Perturbation-based explanations alter input features to evaluate output changes [82]. Gradient-based methods compute token importance using gradient vectors [43]. Surrogate models simplify models while maintaining performance [59]. Decomposition-based methods dissect reasoning into explainable parts [59]. Attention-based methods measure relevance by improving contextual understanding of each input feature [68].

Sample-based Methods. Sample-based methods analyze how output changes result from input variations, focusing on output behavior rather than relevance [59]. Adversarial samples simulate model attacks and expose vulnerable model parts [59], while counterfactual explanations identify minimal input changes needed to alter a model's output [59]. Contrastive explanations investigate why a model predicts x instead of x' [80].

Probing-based Methods. Probing-based methods capture a model's internal knowledge representation [43, 59]. Knowledge probing trains models to identify knowledge insights like syntax or semantic knowledge [82], while concept probing extracts human-level concepts rather than input-specific knowledge [31]. Neuron activation explains neurons of ML models by alternating inputs, a technique used to improve LLMs [51].

Interpretability Methods. Interpretability methods decode model components into human-understandable algorithms [50]. Circuit discovery observes model behavior and conducts experiments to prune model components [12], while causal tracing assesses the impact of neuron activation on a model output [44].

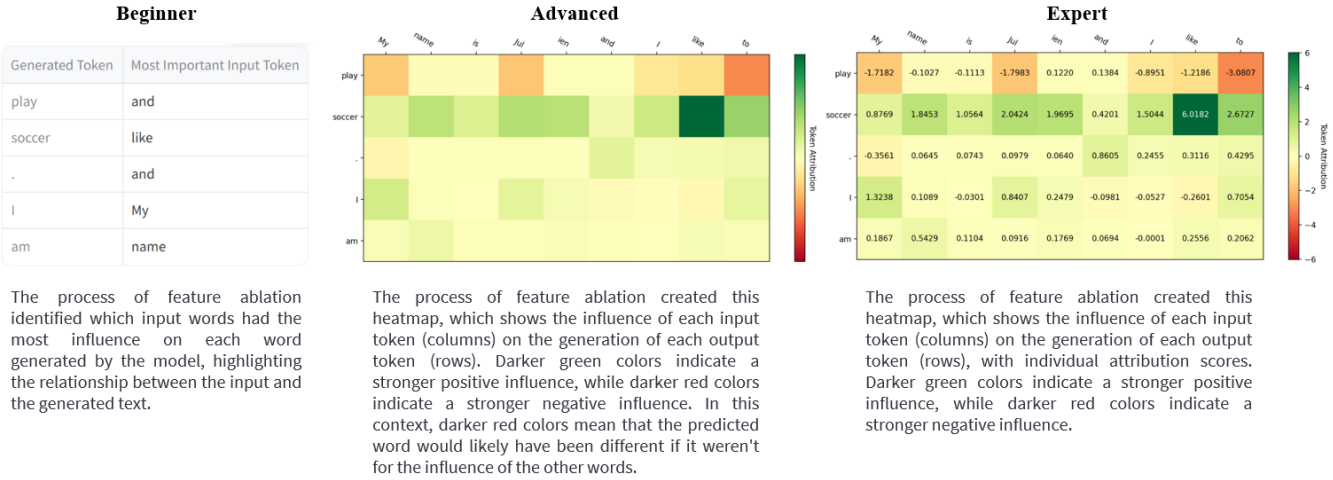
Self-Explanation Methods. LLMs are known for their ability to self-generate explanations for model predictions [77]. Fine-tuning adapts pre-trained models on specific datasets, such as follow-up questions [10], while in-context explanations use the information in a given prompt, like chain-of-thought reasoning [73].

2.3 Selection of Explanation Methods

To allow explainability for all LLMs and be independent of model performance, we needed to focus on post-training explanations. Therefore, probing-based methods and interpretability methods were infeasible. To allow for a variety of explanation methods, we selected one approach of each of the three remaining categories. We chose feature ablation (feature attribution), counterfactuals (sample-based), and self-generated explanations (self-explanation) as explanation methods because they fulfill many of the previously identified personalization techniques (see Table 1).

Feature Ablation. The perturbation-based method *feature ablation* computes the influence of input features by replacing them with a reference and observing the output changes [17, 46, 78]. Feature ablation highlights significant features, demonstrating how their absence or alteration affects the prediction. We chose feature ablation because its approach to creating an explanation is easy to communicate to the end user, regardless of the end user's XAI knowledge level. Its model-agnostic nature allows adaptation to various ML models and representations enabling visualization (like heatmaps) and enhancing accessibility to non-expert users [46]. Therefore, feature ablation fulfills R2 and R6.

Counterfactuals. The sample-based technique *counterfactuals* demonstrates how alternative input data can modify the model's



The process of feature ablation identified which input words had the most influence on each word generated by the model, highlighting the relationship between the input and the generated text.

The process of feature ablation created this heatmap, which shows the influence of each input token (columns) on the generation of each output token (rows). Darker green colors indicate a stronger positive influence, while darker red colors indicate a stronger negative influence. In this context, darker red colors mean that the predicted word would likely have been different if it weren't for the influence of the other words.

The process of feature ablation created this heatmap, which shows the influence of each input token (columns) on the generation of each output token (rows), with individual attribution scores. Darker green colors indicate a stronger positive influence, while darker red colors indicate a stronger negative influence.

Figure 1: Explanations for Text Generation - All Knowledge Levels

output. We adopted Bhattacharjee et al.'s [4] three-step counterfactual generation approach for its clarity and reproducibility. Counterfactuals provide selective explanations by focusing on a few critical input feature changes, thereby creating simpler explanations supporting users in quickly understanding the complex model decision. Counterfactual explanations also support multiple possible explanations for a given output, offering diverse perspectives on outcome achievement. With that, counterfactuals fulfill R5 and R6.

Self-Explainability. Self-explainability of large language models enhances user comprehension by providing natural language explanations, which are more accessible and easier to understand for nontechnical users than complex formulas. This approach allows the incorporation of user knowledge and linking to mental models through model prompts, as well as interactive dialogues and explanations on demand through their dialogue capabilities. Additionally, leveraging state-of-the-art LLMs for dialogues allows the provision of detailed explanations. This approach therefore fulfills R1, R3, and R4.

3 Artifact Implementation

The implementation process involved developing the interactive user interface and its components for personalized explanations. One software developer was responsible for building the prototype, ensuring its functionality and iterative improvements. The following sections highlight the key features of the prototype and its design iterations.

3.1 Interactive User Interface

Profile Settings Tab. To fulfill R1, we implemented a profile settings tab with three components: (1) a toggle button to enable personalized explanations, (2) radio buttons to specify the users knowledge about XAI methods (i.e., *beginner*, *advanced*, *expert*) and (3) checkboxes to select preferred explanation methods, with tool tips for guidance. This design ensures a structured separation of required information types.

Home Tab. To address the rest of the requirements, we implemented a second tab, where predictions and explanations are displayed. The home tab underwent three iterations to optimize the user experience. Each version included an inference area with an input field for text and predictions, two XAI method areas dedicated to displaying feature ablation and counterfactual explanations, and a chat area for interactive, dialogue-based exploration of explanations. Finally, to align with R2, model inference and explanations are grouped in a single row for easier understanding and visual connection of the model prediction and its reasoning. XAI explanations are displayed side by side for better comparison, while the chat is placed below, in a new row, allowing unlimited interaction through scrollable conversations.

3.2 Model Inference

Our prototype was designed around text generation as an LLM use case. Text generation—the automated production of coherent and meaningful text [18]—is a fundamental application of generative AI, widely used in dialogue systems, machine translation, and text summarization [39].

We utilized two different models for text generation tasks: GPT-2, a widely used model with 137 million parameters [54], and Phi-2, a state-of-the-art model with 2.7 billion parameters that offers higher accuracy but requires more computational resources [1]. While Phi-2 is less popular than GPT-2, with approximately 400,000 monthly downloads [25], it enables users to balance the trade-offs between faster inference and improved accuracy.

3.3 Explanation Methods

Each explanation, adjusted to user knowledge level and XAI method, provides the right amount of information without overwhelming the user with details.

Feature Ablation Explanations. Feature ablation identifies the most influential input tokens for a model output and uses attribution scores to indicate their relative importance. In text generation, beginners are provided with input-output tables to map which

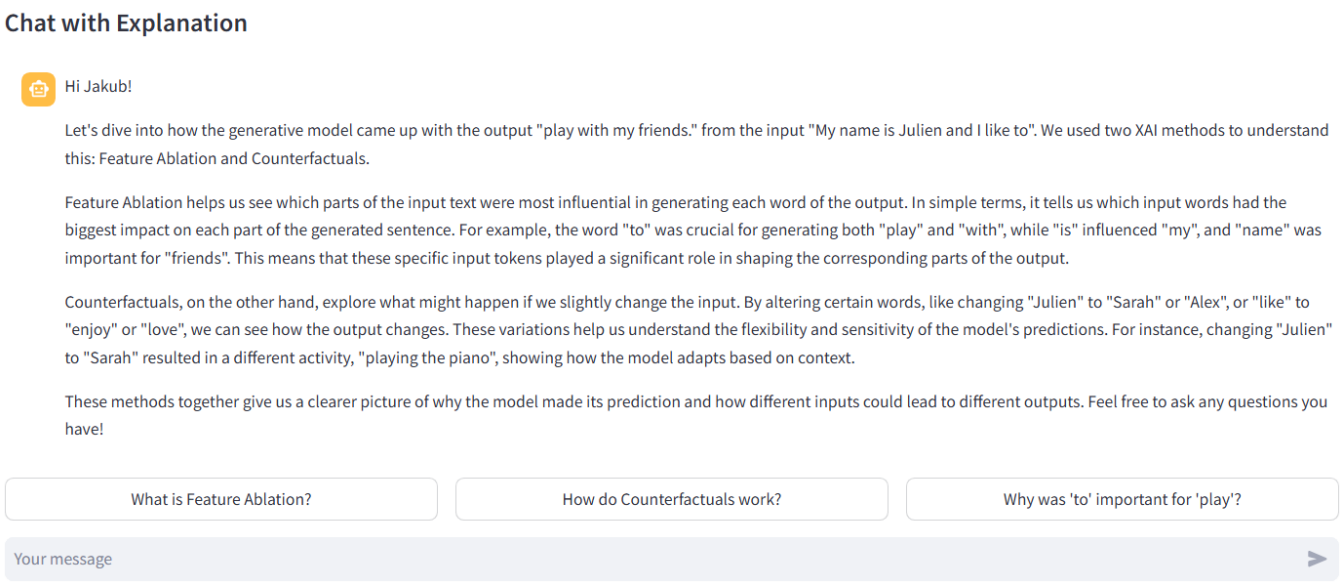


Figure 2: Self-Explainability for a Beginner User

inputs have the greatest influence on specific outputs, advanced users see heatmaps showing overall word influence, and experts receive detailed attribution scores for precise analysis. Figure 1 shows explanations for all three knowledge levels on a sample text generation output.

Counterfactual Explanations. Counterfactual explanations show how input changes affect predictions. In text generation, beginners receive three alternative inputs with highlighted changes, while advanced and expert users receive detailed input components driving these changes.

Self-Explainability Explanations. When self-explainability is enabled, users can interact with feature ablation and counterfactual explanations in a chat-based environment inspired by ChatGPT. The chat environment suggests three possible follow-up questions, based on the user's knowledge level and previous interaction with the chat, helping in understanding the model's behavior. Example follow-up questions for an advanced user are: (1) How do positive attributions affect the output? (2) Can you explain more about counterfactuals? (3) What insights can we gain from this analysis? Users can also ask their own questions via the input box. Figure 2 visualizes a sample chat for a beginner user.

4 Preliminary Evaluation

This section covers a preliminary evaluation conducted with six participants who interacted with TELL-ME for an hour. During this session, they generated predictions, received personalized explanations, and explored the available XAI methods across all three knowledge levels. Additionally, they interacted with the chat feature to ask clarifying questions. They then completed a TAM-based questionnaire [13] on perceived usefulness, ease of use, and overall satisfaction (see Appendix D for details). The participants, aged 22-24 from diverse majors (e.g., geography, mechatronics, computer science), had limited prior XAI experience and considered themselves as beginners in using XAI techniques.

TELL-ME received highly positive feedback, with participants perceiving the prototype as user-friendly, intuitive, and easy to use. They appreciated the interface's simplicity, with comments like "logical and easy design of the user interface" and "very easy design of the interface". Similarly, explanations were deemed easy to understand and interpretable. The participants were satisfied with their overall experience and valued the usefulness of TELL-ME. One participant noted, "I like that you can interact with the prototype and ask questions if you don't fully understand something". Regarding perceived usefulness, most participants found TELL-ME effective in enhancing their understanding of the AI model's prediction process and considered the explanations highly relevant to their needs. Lastly, all participants expressed a behavioral intention to use TELL-ME to understand GenAI outputs in the future, valuing the ability to interact with the prototype as one participant remarked: "I like that I get suggestions for questions to ask in the chat".

However, the evaluation also revealed several challenges of TELL-ME. Feature ablation and textual self-explanation were well-received, but counterfactual explanations were perceived as "not very useful for my understanding of the prediction". Only three participants found counterfactuals to be understandable. This suggests counterfactuals are less helpful, especially for text generation and sentiment analysis. However, counterfactuals might serve as beneficial for expert users requiring additional evaluation. Participants also criticized TELL-ME's limitation to English-only and suggested supporting models like GPT-4o for predictions. However, as GPT-4o is closed-source, its inclusion would require the OpenAI API [52], which would render the current feature ablation approach infeasible. Another concern raised was the token limit

for text generation, which was implemented to prevent visually unappealing explanations, like cluttered heatmaps. Separately, a participant highlighted that "some predictions are sexist", pointing to role assignments based on perceived gender. Additionally, participants suggested adding more prediction types and XAI methods, which is technically feasible. Lastly, one participant suggested further expanding interactivity by making "other explanations more interactive, so I can directly modify parts of the explanation".

5 Discussion and Conclusion

Principal Findings. TELL-ME provides personalizations on three explanation methods (i.e., feature ablation, counterfactuals, and self-explainability) for GenAI outputs on the inference type text generation. Personalization of an explanation itself incorporates six approaches (i.e., user background incorporation, presentation format, interactivity, explanations on demand, complexity reduction, and feature prioritization) for three knowledge levels (i.e., beginner, advanced, and expert) that reflect the user's familiarity with XAI. The interactive chat element of TELL-ME positively improves the overall personalization experience for dynamic, interactive, and personalized explanations [59]. A preliminary evaluation with six participants yielded promising results, demonstrating that the selected explanations and personalizations enhance users' understanding of the model. Participants expressed interest in integrating the prototype into an existing tool, such as ChatGPT, to ease usability and suggested extending support for static explanations, such as heatmaps, and broader language and model support.

Recent research has explored sentiment analysis as a generative AI task, leveraging large language models (LLMs) to improve performance [81]. While our prototype primarily focused on text generation, we also tested sentiment analysis as an additional inference type in the prototype due to its broad applicability in fields such as medicine and social media [71]. Our evaluation results indicate that the prototype also improved users' overall understanding of LLM predictions for this type of inference.

Implications. TELL-ME provides a model-agnostic tool for personalized explanations, addressing the previously stated issue that the "suitability of explanations is question-dependent" [40]. Our prototype is currently focused on text-based tasks but could be extended to other GenAI use cases, such as image and video generation, as well. Its personalized explanations, tailored to user backgrounds, interactive presentation formats, and on-demand delivery, enable easy adaptation across output types (e.g., [60]). Using LLMs for self-explanation offers a novel and effective approach to understand a model's reasoning, especially when combined with supplementary XAI outputs, showcasing their role in advancing user-friendly XAI. TELL-ME's personalization techniques could also be used to adapt explanations for multi-agent systems, collaborative AI applications, or interactive tutoring systems, providing actionable insights tailored to diverse user requirements.

Limitations and Future Research. One of the limitations of this study is the restriction of our initial inference experiments to DistilBERT, GPT-2, and Phi-2 due to availability and computational constraints. Future research could explore personalization with more state-of-the-art models, such as GPT-4o, to generate more accurate and contextually relevant explanations. TELL-ME could also be extended to other GenAI domains, such as image, audio, or video generation. Another limitation regards the self-explainability by GPT-4 [52] that inherits issues of general-purpose LLMs [74]. Although we saw good explanation performance of TELL-ME for text generation and sentiment analysis, hallucinated explanations in high-stakes contexts–such as healthcare, finance, or education–pose significant risks. Adjusting personalization approaches to such domains (e.g., through knowledge inclusion) could help to mitigate inaccuracies and biases. Finally, the evaluation was conducted with a limited sample of university students aged 22–24, which does not represent the broader population. We aim to expand the evaluation to include diverse demographic groups that offer more comprehensive insights and enhance the usability and effectiveness of the prototype.

Conclusion. In conclusion, this paper presents TELL-ME, a prototype that personalizes GenAI explanations using three explanation methods for text generation and sentiment analysis. The subsequent user evaluation indicated that the provided personalized explanations supported the end users in understanding the generative model's reasoning process. Overall, this work contributes to the growing field of XAI for GenAI models by providing a practical approach to personalization.

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A Key Words and Inclusion Criteria for Literature Reviews

A.1 Explanation Personalization

A.1.1 Search String.

("personalize" OR "personalization" OR "customize" OR "customization" OR "human centered") AND "ai" AND ("explanation" OR "explain")
Filtered by publication date: 2020-2024

A.1.2 Inclusion Criteria.

(1) Articles discussing approaches to personalizing explanations for the end user
(2) Articles exploring challenges associated with personalizing explanations in XAI
(3) Studies presenting empirical data on the effectiveness of personalized explanations
(4) Articles including specific examples or case studies illustrating the application of personalized explanations

A.2 Explanation Methods

A.2.1 Search String.

("explainable AI" OR "interpretable AI" OR "XAI")
Filtered by publication date: 2018-2024

A.2.2 Inclusion Criteria.

(1) Articles discussing the methods of XAI
(2) Articles exploring challenges associated with XAI methods
(3) Studies presenting empirical data on the usefulness of XAI methods
(4) Articles including specific examples or case studies illustrating the application of XAI methods

B Concept MatriX for Personalization Approaches

Table 2: Concept Matrix – Personalization Approaches

Article	User Knowledge Incorporation	Interactivity	Complexity Reduction	Feature Prioritization	Presentation Format	Linking to Mental Models	Explanation on Demand
[48]			X	X	X		
[57]			X	X			
[9]				X	X		
[37]	X				X	X	X
[21]		X			X		X
[40]		X					
[45]		X					
[36]	X		X		X	X	
[28]	X						
[33]	X				X		
[49]						X	
[7]							X
[2]		X			X	X	
[38]				X			
[67]		X	X				X
[65]		X					
[35]	X						
[66]	X				X		
[58]	X						
[62]		X					X
[32]			X		X		
[75]					X		
[20]						X	
[11]		X			X		X

C Concept Matrices for Explanation Methods

Table 3: Concept Matrix – XAI Methods – Feature Attributions Methods

Article	Perturbation-based methods	Gradient-based methods	Surrogate Models	Decomposition-based methods	Attention-based methods
[78]	X				
[17]	X	X			
[46]	X	X			
[14]			X		
[61]			X		
[56]			X		
[42]	X		X		
[34]	X			X	
[3]	X				X
[47]		X		X	
[69]					X
[64]					X

Table 4: Concept Matrix – XAI Methods – Sample-Based Methods

Article	Adversarial Samples	Counterfactual Explanations	Contrastive Explanations
[30]	X		
[70]	X		
[76]		X	
[27]		X	
[4]		X	
[80]			X

Table 5: Concept Matrix – XAI Methods – Probing-Based Methods

Article	Knowledge Probing	Concept Probing	Neuron Activation Explanation
[8]	X		
[22]	X		
[82]		X	
[31]		X	
[51]			X

Table 6: Concept Matrix – XAI Methods – Mechanistic Interpretability Methods

Article	Circuit Based Analysis	Causal Tracing
[70]	X	
[12]	X	
[44]		X

Table 7: Concept Matrix – XAI Methods – Self-Explanation Methods

Article	Finetuning Approach	In-Context Approach
[19]	X	
[10]	X	
[23]		X
[73]		X

D Evaluation Questionnaire

Table 8: Personal Information - Results

Attribute	Variable	Amount
Age	22	3
	23	2
	24	1
Course of study	Geography	1
	Mechatronics	1
	Industrial Engineering	1
	Mechanical Engineering	1
	Computer Science	1
	Information Systems	1
Familiarity with XAI	I don't know what XAI is.	2
	I have heard about XAI, but I have not used it yet.	3
	I have seen the application of XAI methods or applied them myself.	1
	I am comfortable in using XAI methods and techniques.	0
	I am an expert when it comes to XAI.	0
Frequency of XAI usage	Not at all.	5
	At least once a year.	0
	At least once a month.	1
	At least once a week.	0

Table 9: Perceived Usefulness (TAM) - Responses

Statement	Variable	Amount
Using the tool enhances my effectiveness in understanding how the AI model makes predictions.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	2
	Strongly Agree	4
Using the tool increases my confidence in the AI model's predictions.	Strongly Disagree	0
	Disagree	1
	Neutral	0
	Agree	5
	Strongly Agree	0
The explanations generated by the tool are relevant to my needs.	Strongly Disagree	0
	Disagree	0
	Neutral	1
The tool helps me identify and correct errors in the AI model predictions.	Strongly Disagree	0
	Disagree	1
	Neutral	1
	Agree	3
	Strongly Agree	1

Table 10: Perceived Ease of Use (TAM) - Responses

Statement	Variable	Amount
I find the tool to be easy to use.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	2
	Strongly Agree	4
The interaction with the tool is clear and understandable.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	2
	Strongly Agree	4
Interacting with the tool does not require a lot of my mental effort.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	2
	Strongly Agree	4
I find it easy to get the tool to do what I want it to do.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	2
	Strongly Agree	4

Table 11: Perceived Usefulness (TAM) - Responses

Statement	Variable	Amount
I like using the tool.	Strongly Disagree	0
	Disagree	5
	Neutral	0
	Agree	5
	Strongly Agree	1
I think it is worthwhile to use the tool.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	6
	Strongly Agree	0
In my opinion, it is very desirable to use the tool for academic and related purposes.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	4
	Strongly Agree	2
I feel satisfied with my overall experience using the tool.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	6
	Strongly Agree	0

Table 12: Attitude Towards Using (TAM) - Responses

Statement	Variable	Amount
I intend to use the tool in the future.	Strongly Disagree	0
	Disagree	0
	Neutral	2
	Agree	4
	Strongly Agree	0
I intend to use the tool to assist my academic activities.	Strongly Disagree	0
	Disagree	0
	Neutral	2
	Agree	4
	Strongly Agree	0
I would recommend this tool to others.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	6
	Strongly Agree	0
I plan to use the tool whenever I need to understand AI model predictions.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	4
	Strongly Agree	2

Table 13: Usability - Responses

Statement	Variable	Amount
The steps to generate explanations are clear and straightforward.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	3
	Strongly Agree	3
I can easily customize the profile settings to suit my preferences.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	0
	Strongly Agree	6
The tool's features are logically organized and easy to find.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	3
	Strongly Agree	3
I can easily understand and interpret the visualizations of the explanations (e.g., heatmaps, counterfactuals).	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	1
	Strongly Agree	5

Table 14: XAI Types and Choices - Responses

Statement	Variable	Amount
I can easily understand and interpret the explanations provided through feature ablation.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	4
	Strongly Agree	2
I can easily understand and interpret the explanations provided through counterfactuals.	Strongly Disagree	0
	Disagree	1
	Neutral	2
	Agree	2
	Strongly Agree	1
I can easily understand and interpret the textual explanations.	Strongly Disagree	0
	Disagree	0
	Neutral	0
	Agree	2
	Strongly Agree	4
When it comes to textual explanations, do you prefer to …	… receive an initial explanation automatically after making a prediction.	6
	… receive explanations after you specifically ask.	0

Table 15: Open Questions - Responses

Question	Answers
What do you like most about the prototype?	1. The heatmaps for visualization, that I can choose different types of model inferences and that I can customize the explanations. 2. The design of the user interface is intuitive and that I can chat with the explanations. 3. Logical and easy design of the user interface. 4. I like that you can interact with the prototype and ask questions if you don't fully understand something. 5. I like that I get suggestions for questions to ask in the chat. 6. Very easy design of the interface.
What do you like least about the prototype?	1. That it's only in English, and not more models are supported. 2. That it does not support the latest models such as GPT-4o for predictions. 3. Counterfactuals are not very useful for my understanding of the prediction. 4. It is in English :(text generation is limited to only 5 words. Some predictions are sexist. 5. I don't like the icon for the "robot" in the chat. 6. More options for types of prediction and explanations would be nice.
Do you have any suggestions for improving the usability or functionality of the tool?	1. Add more supported languages and models. 2. Maybe make the other explanations more interactive, so I can directly modify parts of the explanation and then see how the prediction changes (for the feature ablation explanations in text generation for example). 3. Add more general information about XAI. 4. Support German language & be able to predict more than 5 words (tokens). 5. Redesign the icons for the "robot" and "user" in the chat. Add option to regenerate the suggested questions or print out more possible questions. 6. Add more model inference types.
Would you rather have these functionalities in a separate tool, or would you prefer integrating them into an existing tool (like the user interface for ChatGPT) and why?	1. I think a separate tool wouldn't be as attractive to use as just combining the tool with ChatGPT, as most people already use ChatGPT in their daily lives. 2. Integrated into ChatGPT, because I don't want another new program. 3. I would prefer to have the tool integrated into existing tools like ChatGPT, because it would be easier to use and faster. 4. Integrated in ChatGPT would make it easier for me to use the tool in my daily tasks because I am already using ChatGPT. 5. Included in ChatGPT would make it easier to use. 6. Integrated into an existing tool, because I don't want to switch between different tools.

Received 23 January 2025; accepted 22 February 2025